Week 5 — Research Lab Handouts

Software Quality Models & Quality Management

Software Engineering • Atlantic Technological University (ATU) Donegal

Handout A: ISO/IEC 25010 Reference Sheet

REFERENCE SHEET

Use this sheet to guide your scoring of each characteristic in Task 1. For each characteristic, ask the question shown and look for concrete evidence rather than a gut-feel score.

Characteristic	Guiding Question	Example Evidence to Look For
Functional Suitability	Does the software provide the functions users actually need, correctly and completely?	Missing features; features that produce wrong results; workarounds users have invented
Performance Efficiency	Does it use time and resources appropriately under expected load?	Slow page loads; high memory/CPU use; degradation under many simultaneous users
Compatibility	Can it exchange information with other systems and coexist without conflict?	Failed integrations; data format mismatches; conflicts with other installed software
Usability	Can users understand, learn, operate and find the software appealing to use?	Confusing navigation; unclear error messages; inconsistent UI patterns
Reliability	Does it perform its functions under stated conditions for a stated period without failure?	Crashes; data loss; inconsistent behaviour on repeated identical actions
Security	Does it protect information and data appropriately for the level of access users should have?	Unhashed passwords; missing input validation; excessive permissions granted by default
Maintainability	Can it be effectively modified, corrected or adapted by developers over time?	Duplicated code; poor naming; lack of tests; tightly coupled modules
Portability	Can it be effectively transferred to different environments or platforms?	Hard-coded environment paths; platform-specific dependencies; difficult installation

Score each characteristic from 1 (poor) to 5 (excellent) and write down the specific evidence that justifies your score.

Edition note: this sheet uses the eight characteristics of the 2011 edition of ISO/IEC 25010, which most textbooks still describe. The 2023 revision adds a ninth characteristic, Safety, and renames Usability as Interaction capability and Portability as Flexibility. State which edition you are using in your answers.

Handout B: QA or QC? Activity Cards

TASK 3 CARDS

Quality assurance (QA) is process-oriented and preventive: it defines the standards, procedures and methods that build quality in, and checks the process is followed. Quality control (QC) is product-oriented and detective: it checks specific deliverables against the standards and requirements to find defects. Sort each activity into QA or QC and give a one-line reason. A few are deliberately borderline: when you place one, state the definition you used.

#	Activity	QA or QC?	One-line reason
1	Writing a coding standard for the team	QC	To do with the process of creating a system
2	Running the unit test suite before a change is merged	QA	To do with improving the actual produt; finding the defects
3	Auditing whether teams actually follow the review procedure	QA	To do with ensuring a proper process is being followed
4	Peer review of one pull request	QC	To do with finding any bugs or defects in code
5	Defining the steps of the change-control procedure	QA	To do with the process = procedures
6	Running a static-analysis tool over the code base	QC	To do with the quality of the code
7	Training new starters in the team's testing approach	QA	To do with the teaching a team a procedure
8	User acceptance testing of the finished feature	QC	Finding any bugs or defects in final finished feature
9	Deciding which quality characteristics matter most in the quality plan	QA	The process of picking out most important characteristics
10	Inspecting a design document against the team's template	QC	Inspecting document to detect anything different or out of place
11	Collecting defect counts and reviewing them to change how the team works	QA	A process as the defect have already been found and are being used to fix the process of the team working together
12	Penetration testing the login page	QC	Testing the login page to detect if everything is working as it should be

Handout C: Product and Process Standards — Examples

EXAMPLES TO ADAPT

A product standard applies to the thing being produced. A process standard defines how an activity is carried out. Use these as starting points for Task 4, and adapt them to your own application and team.

Type	Example standard	Characteristics it can protect
Product	Naming conventions and code layout agreed and applied across the code base	Maintainability
Product	A standard template for design documents and README files	Maintainability, Usability (for developers)
Product	A fixed format for error messages and API responses	Compatibility, Usability
Process	Every change is peer reviewed by someone other than its author before merge	Reliability, Security
Process	Automated tests must pass before a change is merged	Reliability, Functional Suitability
Process	Defects are logged, triaged weekly and reviewed for patterns	Reliability, Maintainability